AIR CANVAS: REAL-TIME VIRTUAL DRAWING USING HAND MOTION

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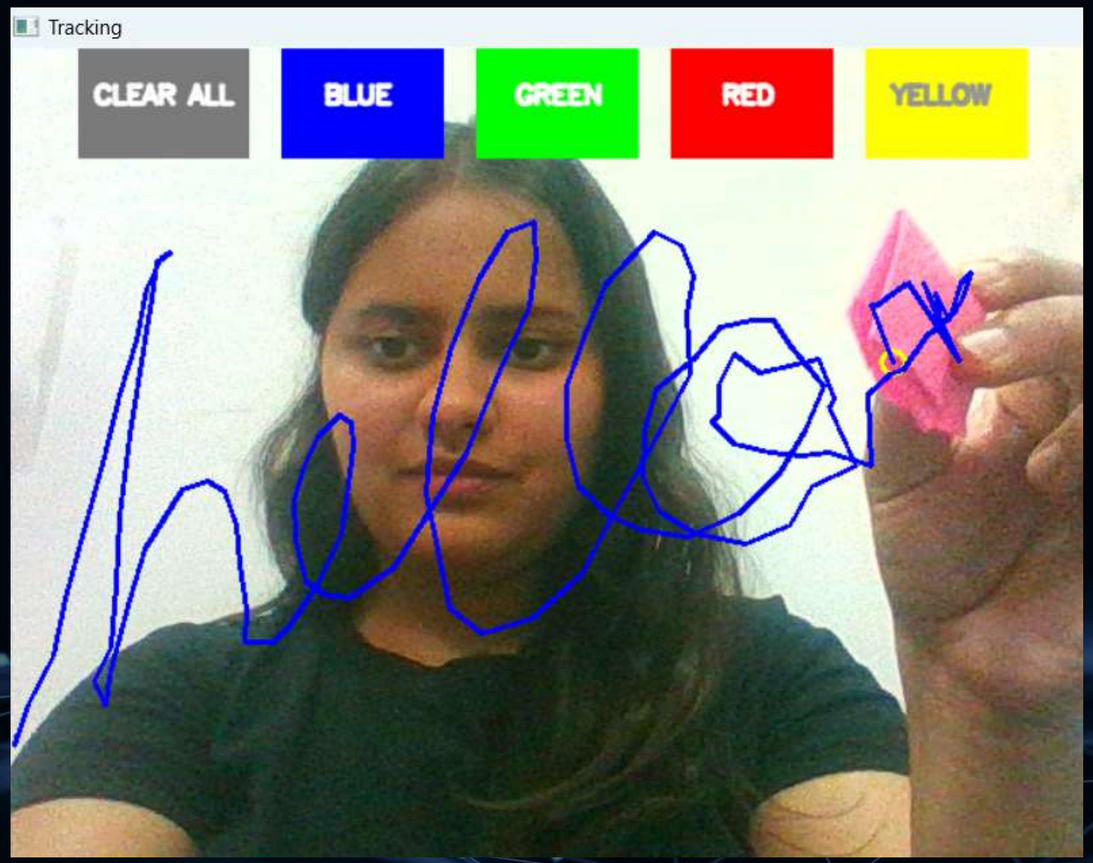
PROBLEM STATEMENT

- With the rise of contactless interaction, there is a need for intuitive drawing methods that don't rely on physical tools.
- This project solves that problem by allowing users to draw in real-time using only a webcam and a colored object.
- It demonstrates the power of computer vision in creating interactive virtual tools.

OBJECTIVE



- To design and implement a real-time virtual canvas using computer vision.
- Use OpenCV and Python to detect a colored object (like a pen cap) in webca feed.
- Provide interactive controls like color selection and canvas clearing.
- Allow users to draw on screen by simply moving that object in the air.

TOOLS AND LIBRARIES UESD

PYTHON 3.X: Programming language used to build the application.

OPENCV: Open-source computer vision library for image processing and webcam access.

NUMPY: Library for handling arrays and matrix operations.

WEBCAM: Input device for capturing real-time video frames.

HOW IT WORKS (STEP-BY-STEP FLOW)

1. Video Capture:

Webcam captures real-time video frames.

2. Colour Conversion:

Each frame is converted from BGR to HSV colour space.

3.SV Range Detection:

The user adjusts the trackbars to define the colour range of their marker (e.g., a pink pen cap).

4.Contour Detection:

The app detects the largest contour (marker) and identifies its centre.

5.Action Trigger:

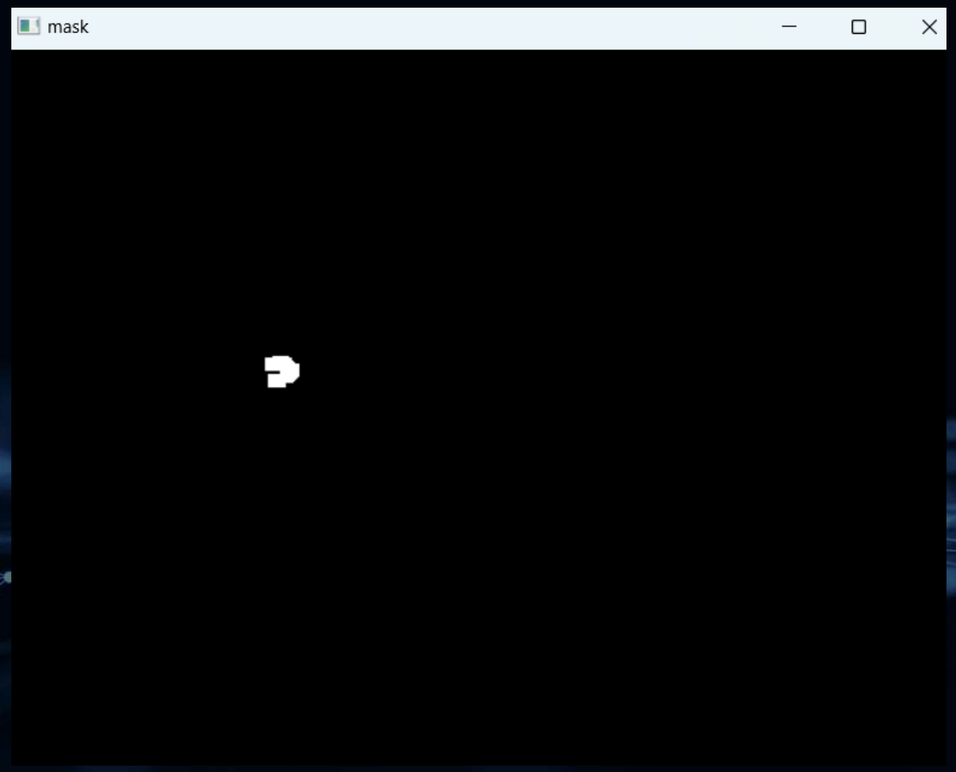
If the marker is over the colour buttons (top bar), it switches colours or clears the canvas.

6.Drawing:

As the marker moves, the system draws lines connecting racked points.

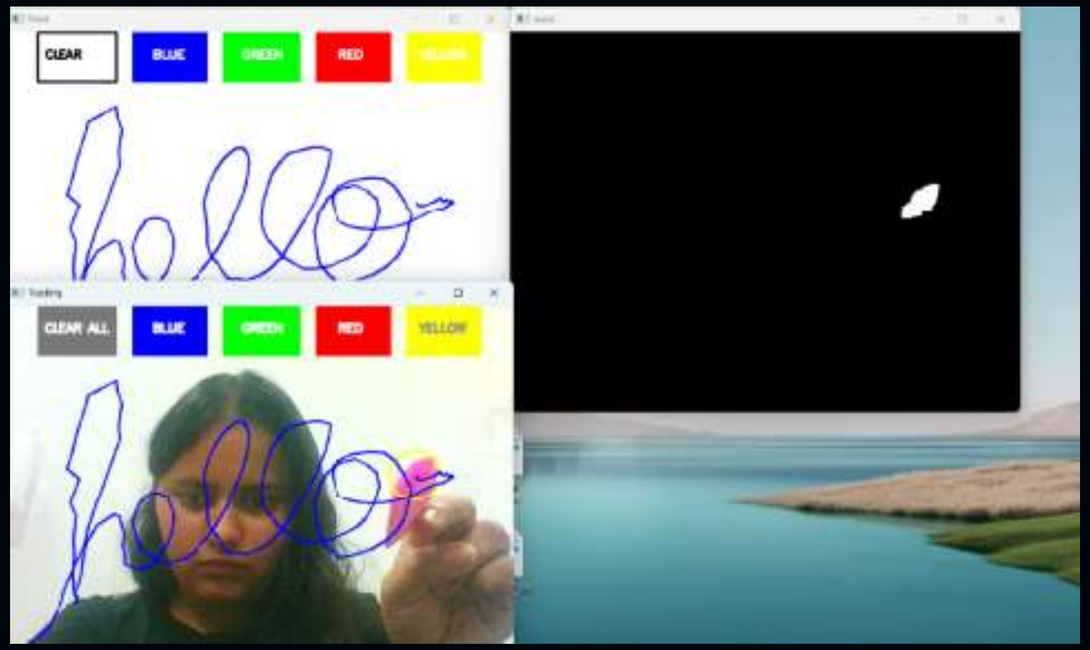
HSV DETECTION IN ACTION

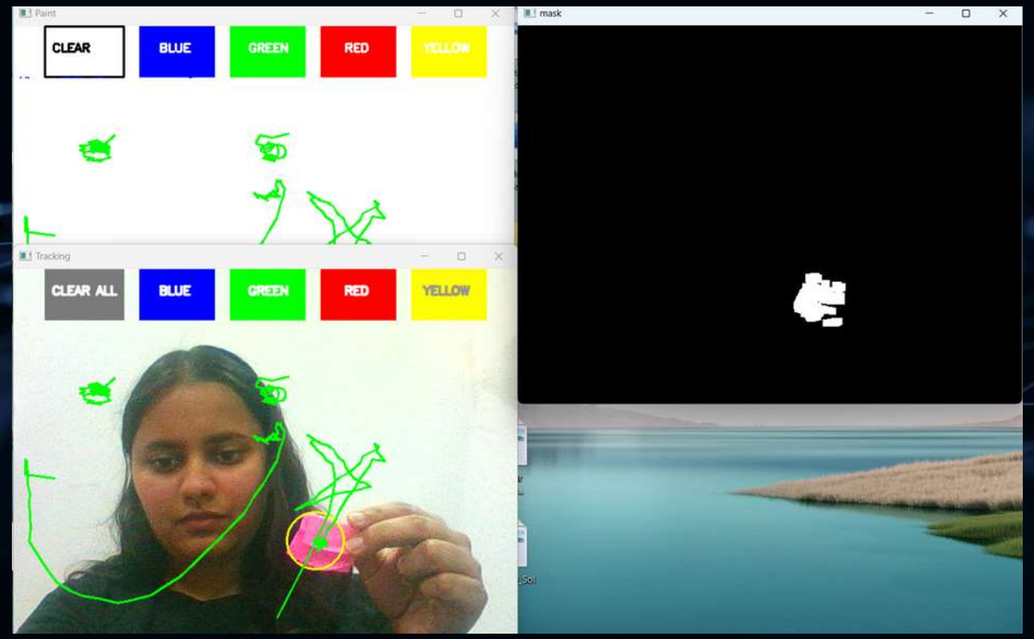
1. HSV Trackbars allow live tuning of detection range:
   - Hue: colour type
   - Saturation: intensity
   - Value: brightness
2. Binary Mask Window:
   - Shows what is currently being detected (white = detected object)

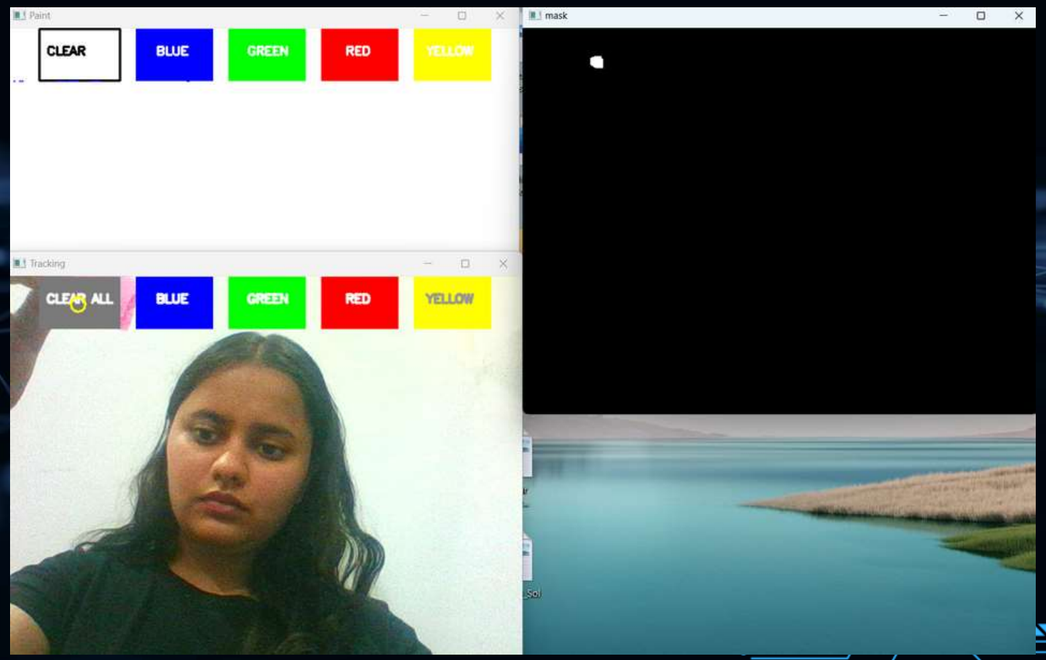


"Proper HSV tuning is critical to ensure the system detects only the intended object and not the background or hair."

DRAWING OUTPUT







APPLICATIONS

1. Touchless Interaction Systems
    - Useful in hygiene-sensitive environments (e.g., hospitals)
2. Educational Tools
    - Fun and engaging tech for kids to draw and learn
3. Gesture-Controlled Interfaces
    - Prototype for future AR/VR and smart TV control
4. Rehabilitation & Therapy
    - Could be adapted for motor skill training

CHALLENGES & FUTURE SCOPE

Challenges Faced:

1. HSV tuning sensitivity to lighting
2. False detection of face/hair if background was similar in color
3. Webcam latency and flickering

Future Improvements:

1. Add fingertip detection using MediaPipe
2. Export drawings as images
3. Add gesture-based commands (e.g., wave = undo)
4. Integrate voice commands for interaction

THANK YOU

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